

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20										820	ALLOY 825 INTERNAL CLADDING IS
20		VA5					***** MODULAR VALVES *****				ACCEPTABLE FOR PIPES MANUFACTURED
30			MODULAR DBB	K03AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		FROM METALLURGICALLY ROLL-BONDED OR
40		VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		EXPLOSION BONDED CLAD PLATES. ALLOY 625
50			MODULAR DBB	M03AM10	0.5	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		MATERIAL TO BE USED FOR PIPES
60		VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	B1	MANUFACTURED WITH INTERNAL WELD
70		VAW	MODULAR DBB	M03EM10	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	B6	OVERLAY.
80	U04	VAW	MODULAR DBB	M03EM10L	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	FA8	ALL BOLTS ARE METRIC DIA.
90	U04	VAW	MODULAR DBB	M03EM10L	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	FF5	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1
100											INCH 1500# PERAR BALL VALVES ITEM CODE
100							***** BALL VALVES *****				B15AS22
110			BALL	A03AM10	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING METAL RB LO	0	0		USE THREADED FLANGES FOR INSTRUMENT
120			BALL	A03AR10	4	4	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM FLOATING METAL RB LO	0	0		CONNECTIONS ONLY.
130			BALL	A03AR16	6	8	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION METAL RB GO	0	0		FLAT BOTTOM NIPOF LANGES FOR USE ON
140	U04		BALL	B03AM10	0.75	2	BALL 300# RF INC 825 INC 625 TRIM FLOATING METAL FB LO	0	0		WELD CAPS AND BLIND FLANGES
150	U04	VA5	BALL	B03AM10	3	3	BALL 300# RF INC 825 INC 625 TRIM FLOATING METAL FB LO	0	0		USE FLANGES, GASKETS & VALVES AS
160	U04	VA5	BALL	B03AR12	4	4	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING METAL FB GO	0	0		REQUIRED TO MATCH EQUIPMENT OR VESSEL
170	U04	VA5	BALL	B03AR16	6	24	BALL 300# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION METAL FB GO	0	0		NOZZLES.
180							***** BUTTERFLY VALVES *****				SPECIAL GASKET ONLY FOR USE AS SPECIFIED
180											IN ENG-200-117
190			BUTTERFLY	T03AR63	10	24	BUTTERFLY 300# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0		USE ONLY WITH CONSENT OF MATERIALS
200							***** GATE VALVES *****				ENGINEER.
200		VQ2									PS1
210			GATE	G03AM49	0.75	3	GATE 300# RF INC 825 INC 625 TRIM WEDGE HW	0	0		PIPE ONLY TO BE USED AS TRUNNION
220	U04		GATE	W03AL73	8	8	GATE 300# RF INCONEL 625 N26625 THROUGH COND SOLID SLAB GO	0	0		SUPPORTS
230	U04		GATE	W03AM71	8	8	GATE 300# RF INCONEL 825 N08825 THROUGH COND SOLID SLAB GO	0	0		PIPE ONLY TO BE USED AS TRUNNION
240							***** GLOBE VALVES *****				SUPPORTS. USE FOR HEADER SIZE 4" AND ABOVE ONLY.
240											SB6
250											USE JACKSCREWS FOR PADDLE
250			GLOBE	L03AM49	0.75	3	GLOBE 300# RF INC 825 INC 625 TRIM OS&Y HW	0	0		BLANKS/SPACERS AND FLANGE SPREADERS
260			GLOBE	L03AR49	4	6	GLOBE 300# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0		FOR SPECTACLE BLINDS.
270			GLOBE	L03AR44	8	10	GLOBE 300# RF LTCS + 825 CLAD INC 625 TRIM OS&Y GO	0	0		USE 1 1/2IN WITH THERMOWELL
280							***** CHECK VALVES *****				CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR
280											DBB PRESSURE CONNECTIONS WITH
290											CHEMICAL SEALS.
290		VC1	CHECK	C03AM51	0.75	1.5	CHECK PISTON 300# RF INC 825 TRIM 625	0	0		VA5
300	U04		CHECK	C03AM46	2	3	CHECK DUAL WAFER LUG 300# RF INC 825 INC TRIM 625 THROU BOLTED	0	0		USE THESE VALVES ONLY WHERE SPECIFIED
310	U04		CHECK	C03AR46	4	10	CHECK DUAL WAFER LUG 300# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0		ON PIPING AND INSTRUMENT DIAGRAMS
320	U04		CHECK	C03AR53	12	24	CHECK DUAL DBL FLG 300# RF LTCS + CLAD INC 825 TRIM 625	0	0		USAGE IS LIMITED TO INSTRUMENT ISOLATION
330							***** PIPES*****				PURPOSE.
330											VAV
340			PIPE	5732001	0.75	1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE	0	0		VALVE ONLY TO BE USED IN ACCORDANCE
350			PIPE	5361603	1.5	3	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE	0	0		WITH DOCUMENT L-ST-2055 - VENT DRAIN AND
360			PIPE	5358505_C80	4	14	PIPE SCH STD SMLS LTCS A333 GR.6 WITH 3MM INC 825 INTERNAL CLADDING	0	0		INSTRUMENT CONNECTION DETAILS.
370			PIPE	5358514_C80	16	16	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0		VC1
380		810	PIPE	5732003	0.5	0.5	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE	0	0		SPRING PISTON CHECK VALVES ARE PREFERRED
390	U04		PIPE	5358510_C60	4	4	PIPE SCH 120 SMLS LTCS A333 GR.6 BE WITH 3MM INC 625 INTERNAL CLADDING	0	0		IN THE HORIZONTAL POSITION, BUT MAY BE
400	U04	820	PIPE	5358385_C60	24	24	PIPE SCH 30 EFW LTCS A671 CC60 CL.22 WITH 3 MM INC 825/625 INTERNAL CLADDING	0	0		USED IN THE VERTICAL POSITION WITH
410	U04		PIPE	5358506_C60	8	8	PIPE SCH 60 SMLS LTCS A333 GR.6 BE WITH 3 MM INC 625 INTERNAL CLADDING	0	0		ENGINEERING APPROVAL.
420	U04		PIPE	5358505_C60	4	14	PIPE SCH STD SMLS LTCS A333 GR.6 WITH 3MM INC 625 INTERNAL CLADDING	0	0		VQ2
430	U04	820	PIPE	5358577_C60	18	20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 WITH 3 MM INC 825/625 INTERNAL CLADDING	0	0		QUARTER TURN VALVES ARE TO BE SELECTED
440	U04		PIPE	5358514_C60	6	6	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0		FOR PROCESS APPLICATIONS.
450	U04		PIPE	5358514_C60	16	16	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0		
460	U04	820	PIPE	5358578_C60	42	42	PIPE WT 24.0MM EFW LTCS A671 CC60 CL.22 WITH 3MM INC 825/625 INTERNAL CLADDING	0	0		
470							***** TRUNNION PIPE SUPPORTS *****				
470											
480											
480	U04	PS1	PIPE	5358510	4	4	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0		
490	U04	PS1	PIPE	5358385	24	24	PIPE SCH 30 EFW LTCS A671 CC60 CL.22 BE	0	0		
500	U04	PS4	PIPE	5358504	3	3	PIPE SCH 40 SMLS LTCS A333GR.6 BE	0	0		
510	U04	PS1	PIPE	5358506	8	8	PIPE SCH 60 SMLS LTCS A333 GR.6 BE	0	0		
520	U04	PS1	PIPE	5358505	10	12	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZE - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 1 of 5	300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
530	U04	PS1	PIPE	5358577	20	20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE	0	0	
540	U04	PS1	PIPE	5358514	6	6	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0	
550	U04	PS1	PIPE	5358514	16	16	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0	
560							*****FITTINGS*****			
570										
570			ELBOW	5425111	0.75	1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0	
580			ELBOW	5424722	1.5	3	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0	
590			ELBOW	5392438_C80	4	14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0	
600			ELBOW	5392600_C80	16	16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
610			ELBOW	5392497_C80	18	20	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
620			ELBOW	5392331_C80	24	24	ELL 90 DEG LR SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
630			ELBOW	5392551_C80	42	42	ELL 90 DEG LR WT 24.0MM LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	0	0	
640	U04		ELBOW	5392331_C60	24	24	ELL 90 DEG LR SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
650	U04		ELBOW	5392551_C60	42	42	ELL 90 DEG LR WT 24.0MM LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0	
660	U04		ELBOW	5392700_C60	4	4	ELL BW 90 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
670	U04		ELBOW	5392492_C60	8	8	ELL BW 90 DEG LR SCH 60 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
680	U04		ELBOW	5392438_C60	4	14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
690	U04		ELBOW	5392600_C60	6	6	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
700	U04		ELBOW	5392600_C60	16	16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
710	U04		ELBOW	5392497_C60	18	20	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
720			ELBOW	5425113	0.75	1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0	
730			ELBOW	5424727	1.5	3	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0	
740			ELBOW	5392444_C80	4	14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0	
750			ELBOW	5392606_C80	16	16	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
760			ELBOW	5392608_C80	18	20	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
770			ELBOW	5392337_C80	24	24	ELL 45 DEG SCH 30 LTCS A420 WPL6 -W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
780	U04		ELBOW	5392337_C60	24	24	ELL 45 DEG SCH 30 LTCS A420 WPL6 -W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
790	U04		ELBOW	5392498_C60	8	8	ELL BW 45 DEG LR SCH 60 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
800	U04		ELBOW	5392444_C60	4	14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
810	U04		ELBOW	5392606_C60	6	6	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
820	U04		ELBOW	5392606_C60	16	16	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
830	U04		ELBOW	5392608_C60	18	20	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
840										
840			TEE	5425119	0.75	1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0	
850			TEE	5424725	1.5	3	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0	
860			TEE	5392447_C80	4	14	TEE BW SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	0	0	
870			TEE	5392609_C80	16	16	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
880			TEE	5392406_C80	18	20	TEE BW SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
890			TEE	5392412_C80	24	24	TEE SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
900	U04		TEE	5392702_C60	4	4	TEE BW SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
910	U04		TEE	5392501_C60	8	8	TEE BW SCH 60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
920	U04		TEE	5392447_C60	4	14	TEE BW SCH STD LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
930	U04		TEE	5392609_C60	6	6	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
940	U04		TEE	5392609_C60	16	16	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
950	U04		TEE	5392406_C60	18	20	TEE BW SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
960			TEE RED	5425126	0.75	0.75	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1	1	
970			TEE RED	5425124	0.75	1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1.5	2	
980			TEE RED	5424724	1.5	2	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2	3	
990			TEE RED	5425128	1.5	3	TEE RED SCH STD/SCH 40S ALLOY 825 B366-WPS UNS N08825	4	6	
1000			TEE RED	5392445_C80	4	12	TEE RED BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	14	
1010			TEE RED	5392460_C80	6	14	TEE RED BW SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16	16	
1020			TEE RED	5392476_C80	8	14	TEE RED SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18	20	
1030			TEE RED	5393023_C80	10	14	TEE RED SCH 30/ SCH STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24	24	
1040			TEE RED	5393022_C80	16	20	TEE RED SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24	24	
1050	U04		TEE RED	5392427_C60	4	4	TEE RED BW SCH 60/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8	
1060	U04		TEE RED	5425179_C60	6	6	TEE RED BW SCH 60/SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8	
1070	U04		TEE RED	5392445_C60	4	12	TEE RED BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	14	
1080	U04		TEE RED	5392451_C60	4	4	TEE RED BW SCH STD/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	10	
1090	U04		TEE RED	5392348_C60	8	8	TEE RED BW SCH STD/SCH 60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	14	
1100	U04		TEE RED	5392454_C60	6	6	TEE RED BW SCH STD/XS LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	10	14	
1110	U04		TEE RED	5392413_C60	6	6	TEE RED BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16	
1120	U04		TEE RED	5392626_C60	4	4	TEE RED BW SCH XS/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	6	
1130	U04		TEE RED	5392460_C60	10	14	TEE RED BW SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16	
1140	U04		TEE RED	5392471_C60	3	3	TEE RED SCH 120/160 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	4	4	
1150	U04		TEE RED	5392442_C60	2	2	TEE RED SCH 120/XS LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	4	4	
1160	U04		TEE RED	5393023_C60	10	14	TEE RED SCH 30/ SCH STD LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	24	24	
1170	U04		TEE RED	5393022_C60	16	20	TEE RED SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	24	24	
1180	U04		TEE RED	5392469_C60	3	3	TEE RED SCH XS/160 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	6	6	
1190	U04		TEE RED	5392586_C60	8	8	TEE RED SCH XS/60 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	16	16	
1200	U04		TEE RED	5392589_C60	8	8	TEE RED SCH XS/SCH 60 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20	
1210	U04		TEE RED	5392476_C60	8	14	TEE RED SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20	
1220	U04		TEE	5392412_C60	24	24	TEE SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZE - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
1230														
1230			CAP	5424270	0.75	1	CAP SCH 80S ALLOY 825 B366 UNS N08825	0	0					
1240			CAP	5424219	1.5	3	CAP SCH 40S ALLOY 825 B366 UNS N08825	0	0					
1250			CAP	5392448_C80	4	14	CAP BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0					
1260			CAP	5392610_C80	16	20	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0					
1270			CAP	5392340_C80	24	24	CAP SCH 30 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0					
1280	U04		CAP	5392502_C60	8	8	CAP BW SCH 60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
1290	U04		CAP	5392448_C60	4	14	CAP BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
1300	U04		CAP	5392610_C60	6	6	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
1310	U04		CAP	5392610_C60	16	20	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
1320	U04		CAP	5392703_C60	4	4	CAP SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0					
1330	U04		CAP	5392340_C60	24	24	CAP SCH 30 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
1340														
1340			RED CONC	5425122	0.75	0.75	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1	1					
1350			RED CONC	5425143	0.75	1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1.5	2					
1360			RED CONC	5424733	1.5	2	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2	3					
1370			RED CONC	5425146	1.5	3	REDUCER CONC SCH STD/40S ALLOY 825 B366-WPS N08825	4	6					
1380			RED CONC	5392449_C80	4	12	REDUCER CONC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	14					
1390			RED CONC	5392457_C80	8	14	REDUCER CONC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16	16					
1400			RED CONC	5392506_C80	10	14	REDUCER CONC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18	20					
1410			RED CONC	5392507_C80	16	16	REDUCER CONC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18	20					
1420			RED CONC	5392523_C80	16	20	REDUCER CONC SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24	24					
1430	U04		RED CONC	5425161	3	3	REDUCER CONC SCH 120/40S ALLOY 825 B366-WPS N08825	6	6					
1440	U04		RED CONC	5392741	1.5	3	REDUCER CONC SCH 160/40S ALLOY 825 B366-WPS N08825	4	4					
1450	U04		RED CONC	5392523_C60	16	20	REDUCER CONC SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	24	24					
1460	U04		RED CONC	5392343_C60	4	4	REDUCER CONC SCH 60/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8					
1470	U04		RED CONC	5392344_C60	6	6	REDUCER CONC SCH 60/SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8					
1480	U04		RED CONC	5392449_C60	4	12	REDUCER CONC SCH STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	14					
1490	U04		RED CONC	5392599_C60	4	4	REDUCER CONC SCH STD/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	10					
1500	U04		RED CONC	5392629_C60	8	8	REDUCER CONC SCH STD/SCH 60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	14					
1510	U04		RED CONC	5392452_C60	6	6	REDUCER CONC SCH STD/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	14					
1520	U04		RED CONC	5392458_C60	6	6	REDUCER CONC SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1530	U04		RED CONC	5392507_C60	16	16	REDUCER CONC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20					
1540	U04		RED CONC	5392740_C60	4	4	REDUCER CONC SCH XS/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	6					
1550	U04		RED CONC	5392594_C60	8	8	REDUCER CONC SCH XS/60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1560	U04		RED CONC	5392457_C60	8	14	REDUCER CONC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1570	U04		RED CONC	5392506_C60	10	14	REDUCER CONC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20					
1580			RED ECC	5425123	0.75	0.75	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1	1					
1590			RED ECC	5425145	0.75	1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1.5	2					
1600			RED ECC	5424734	1.5	2	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2	3					
1610			RED ECC	5425148	1.5	3	REDUCER ECC SCH STD/40S ALLOY 825 B366-WPS N08825	4	6					
1620			RED ECC	5392450_C80	4	12	REDUCER ECC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	14					
1630			RED ECC	5392459_C80	8	14	REDUCER ECC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16	16					
1640			RED ECC	5392510_C80	10	14	REDUCER ECC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18	20					
1650			RED ECC	5392514_C80	16	18	REDUCER ECC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18	20					
1660			RED ECC	5392397_C80	16	20	REDUCER ECC SCH 30/XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24	24					
1670	U04		RED ECC	5425162	3	3	REDUCER ECC SCH 120/40S ALLOY 825 B366-WPS N08825	6	6					
1680	U04		RED ECC	5425159	1.5	3	REDUCER ECC SCH 160/40S ALLOY 825 B366-WPS N08825	4	4					
1690	U04		RED ECC	5392397_C60	16	20	REDUCER ECC SCH 30/XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	24	24					
1700	U04		RED ECC	5392643_C60	4	4	REDUCER ECC SCH 60/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8					
1710	U04		RED ECC	5392645_C60	6	6	REDUCER ECC SCH 60/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	8	8					
1720	U04		RED ECC	5392450_C60	4	12	REDUCER ECC SCH STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	14					
1730	U04		RED ECC	5392644_C60	4	4	REDUCER ECC SCH STD/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	10					
1740	U04		RED ECC	5392839_C60	8	8	REDUCER ECC SCH STD/SCH 60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	14					
1750	U04		RED ECC	5392453_C60	6	6	REDUCER ECC SCH STD/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	14					
1760	U04		RED ECC	5392461_C60	6	6	REDUCER ECC SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1770	U04		RED ECC	5392514_C60	16	18	REDUCER ECC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20					
1780	U04		RED ECC	5392597_C60	8	8	REDUCER ECC SCH XS/60 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1790	U04		RED ECC	5392820_C60	4	4	REDUCER ECC SCH XS/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	6					
1800	U04		RED ECC	5392459_C60	10	14	REDUCER ECC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	16	16					
1810	U04		RED ECC	5392510_C60	10	14	REDUCER ECC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	18	20					
1820														
1820			OLETS	5513491	0.75	1	NIPOFLANGE 300# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2	3					
1830			OLETS	5513492	0.75	1	NIPOFLANGE 300# RF SCH STD/80S ALLOY 825 B564 UNS N08825	4	14					
1840			OLETS	5513493	0.75	1	NIPOFLANGE 300# RF SCH XS/80S ALLOY 825 B564 UNS N08825	16	20					
1850	U04		OLETS	5513493	0.75	1	NIPOFLANGE 300# RF SCH XS/80S ALLOY 825 B564 UNS N08825	6	6					
1860			OLETS	5513494	0.75	1	NIPOFLANGE 300# RF SCH SCH 30/80S ALLOY 825 B564 UNS N08825	24	24					
1870			OLETS	5513495	1.5	2	NIPOFLANGE 300# RF SCH 40S ALLOY 825 B564 UNS N08825	3	3					
1880			OLETS	5513496	1.5	2	NIPOFLANGE 300# RF SCH STD/ 40S ALLOY 825 B564 UNS N08825	4	14					
1890			OLETS	5513497	1.5	2	NIPOFLANGE 300# RF SCH XS/40S ALLOY 825 B564 UNS N08825	16	20					
1900	U04		OLETS	5513497	1.5	2	NIPOFLANGE 300# RF SCH XS/40S ALLOY 825 B564 UNS N08825	6	6					
1910			OLETS	5513498	1.5	2	NIPOFLANGE 300# RF SCH 30/40S ALLOY 825 B564 UNS N08825	24	24					
1920			OLETS	5513499	1.5	2	NIPOFLANGE 300# RF WT 24.0MM/40S ALLOY 825 B564 UNS N08825	42	42					

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZE - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 3 of 5	300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1930		FF5	OLETS	5504518	0.75	1	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S ALLOY 825 B564 UNS N08825	0	0	
1940		FF5	OLETS	5504519	1.5	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 40S ALLOY 825 B564 UNS N08825	0	0	
1950		UJ2	OLETS	5513479	1.5	1.5	NIPOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4	14	
1960		UJ2	OLETS	5513478	1.5	1.5	NIPOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	16	20	
1970	U04	UJ2	OLETS	5513478	1.5	1.5	NIPOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	6	6	
1980		UJ2	OLETS	5513477	1.5	1.5	NIPOFLANGE 300# RF SCH 30/160 ALLOY 825 B564 UNS N08825	24	24	
1990	U04	UJ2	OLETS	5513579	1.5	1.5	NIPOFLANGE 300# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4	4	
2000	U04		OLETS	5513520	1.5	2	NIPOFLANGE 300# RF SCH 120/40S ALLOY 825 B564 UNS N08825	4	4	
2010	U04		OLETS	5513476	0.75	1	NIPOFLANGE 300# RF SCH 120/80S ALLOY 825 B564 UNS N08825	4	4	
2020	U04	UJ2	OLETS	5513522	1.5	1.5	NIPOFLANGE 300# RF SCH 60/160 ALLOY 825 B564 UNS N08825	8	8	
2030	U04		OLETS	5513521	1.5	2	NIPOFLANGE 300# RF SCH 60/40S ALLOY 825 B564 UNS N08825	8	8	
2040	U04		OLETS	5513502	0.75	1	NIPOFLANGE 300# RF SCH 60/80S ALLOY 825 B564 UNS N08825	8	8	
2050	U04	FS1	OLETS	5513503	2	2	NIPOFLANGE 600# RF SCH STD/40S ALLOY 825 B564 UNS N08825	10	10	
2060										
2060			OLETS	5513413	0.75	1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2	3	
2070			OLETS	5513417	0.75	1	WELDOLET SCH STD/80S ALLOY 825 B564 UNS N08825	4	14	
2080			OLETS	5513415	0.75	1	WELDOLET SCH XS/80S ALLOY 825 B564 UNS N08825	16	20	
2090	U04		OLETS	5513415	0.75	1	WELDOLET SCH XS/80S ALLOY 825 B564 UNS N08825	6	6	
2100			OLETS	5513416	0.75	1	WELDOLET SCH 30/80S ALLOY 825 B564 UNS N08825	24	24	
2110			OLETS	5513418	1.5	3	WELDOLET SCH STD/ 40S ALLOY 825 B564 UNS N08825	4	14	
2120			OLETS	5513419	1.5	3	WELDOLET SCH XS/ 40S ALLOY 825 B564 UNS N08825	16	20	
2130	U04		OLETS	5513419	1.5	3	WELDOLET SCH XS/ 40S ALLOY 825 B564 UNS N08825	6	6	
2140			OLETS	5513410	1.5	3	WELDOLET SCH 30/ 40S ALLOY 825 B564 UNS N08825	24	24	
2150	U04		OLETS	5505164_C80	4	4	WELDOLET SCH STD LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	12	14	
2160	U04		OLETS	5505334_C80	4	6	WELDOLET SCH XS/STD LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	16	20	
2170	U04		OLETS	5505336_C80	4	8	WELDOLET SCH 30/STD LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	24	24	
2180	U04	new	OLETS	5505338_C80	6	6	WELDOLET WT 24.0MM/STD LTCS A350 LF2 CL.1 WITH 3MM INC 826 INTERNAL CLADDING	42	42	
2190	U04		OLETS	5504381_C60	4	4	WELDOLET FLAT BOTTOM SCH 120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
2200	U04		OLETS	5513403	1.5	2	WELDOLET FLAT BOTTOM SCH 40S ALLOY 825 B564 UNS N08825	0	0	
2210	U04		OLETS	5513406	0.75	0.75	WELDOLET FLAT BOTTOM SCH 80S ALLOY 825 B564 UNS N08825	0	0	
2220	U04		OLETS	5509610	1.5	3	WELDOLET SCH 120 /40S ALLOY 825 B564 UNS N08825	4	4	
2230	U04		OLETS	5509620	0.75	1	WELDOLET SCH 120 /80S ALLOY 825 B564 UNS N08825	4	4	
2240	U04		OLETS	5513532_C60	4	4	WELDOLET SCH 30/120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	24	24	
2250	U04		OLETS	5513533_C60	8	8	WELDOLET SCH 30/60 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	24	24	
2260	U04		OLETS	5505573_C60	6	6	WELDOLET SCH 30/XS LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	24	24	
2270	U04		OLETS	5513432	1.5	3	WELDOLET SCH 60/40S ALLOY 825 B564 UNS N08825	8	8	
2280	U04		OLETS	5513433	0.75	1	WELDOLET SCH 60/80S ALLOY 825 B564 UNS N08825	8	8	
2290	U04		OLETS	5513530_C60	4	4	WELDOLET SCH STD/120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	12	14	
2300	U04		OLETS	5505335_C60	6	6	WELDOLET SCH XS LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	16	20	
2310	U04		OLETS	5513531_C60	4	4	WELDOLET SCH XS/120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	16	20	
2320	U04		OLETS	5505595_C60	8	8	WELDOLET WT 24.0MM/ SCH 60 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	42	42	
2330	U04		OLETS	5513534_C60	6	6	WELDOLET WT 24.0MM/XS LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	42	42	
2340										
2340			ELBOFLG	5516626	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4	14	
2350	U04		ELBOFLG	5516627	1.5	1.5	ELBOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	6	6	
2360			ELBOFLG	5516627	1.5	1.5	ELBOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	16	20	
2370			ELBOFLG	5516628	1.5	1.5	ELBOFLANGE 300# RF SCH 30/160 ALLOY 825 B564 UNS N08825	24	24	
2380	U04		ELBOFLG	5516631	1.5	1.5	ELBOFLANGE 300# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4	4	
2390	U04		ELBOFLG	5516632	1.5	1.5	ELBOFLANGE 300# RF SCH 60/160 ALLOY 825 B564 UNS N08825	8	8	
2400										
2400		FS1					*****FLANGES*****			
2410										
2410			FLANGE	5569290	0.75	1	FLG WN 300# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0	
2420			FLANGE	5569276	1.5	3	FLG WN 300# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0	
2430	U04		FLANGE	5556473_C80	4	14	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2440	U04		FLANGE	5556476_C80	16	20	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2450	U04		FLANGE	5556471_C80	24	24	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 30 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2460	U04		FLANGE	5556488_C80	42	42	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 24.0MM B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2470	U04		FLANGE	5556451_C60	4	4	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2480	U04		FLANGE	5556471_C60	24	24	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 30 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2490	U04		FLANGE	5556452_C60	8	8	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 60 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2500	U04		FLANGE	5556473_C60	4	14	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2510	U04		FLANGE	5556476_C60	6	6	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2520	U04		FLANGE	5556476_C60	16	20	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2530	U04		FLANGE	5556488_C60	42	42	FLG WN 300# RF LTCS A350 LF2 CL.1 WT 24.0MM B16.47 SERIES A WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2540	U04	FS1	FLANGE	5569392	1.5	3	FLG WN 600# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0	
2550	U04	FS1	FLANGE	5569391	0.75	1	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0	
2560	U04	FS1	FLANGE	5556969_C60	8	8	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 60 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2570	U04	FS1	FLANGE	5557033_C60	8	10	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2580	U04		FLANGE	5557141_C60	6	6	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2590			BLIND	5617159	0.5	3	FLG BLIND 300# RF ALLOY 825 B564 UNS N08825	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZE - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 4 of 5	300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2600	U04		BLIND	5615576_C80	4	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE	0	0	
2610	U04		BLIND	5615576_C60	4	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE	0	0	
2620	U04	FS1	FLANGE	5557033_C80	8	10	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2630		FA8	TEE RED	5614310	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF ALLOY 825 B564 UNS N08825	0.75	1.5	
2640										
2640			FLANGE	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
2650	U04	ND9	FLG ADAPTER	5851222	0.5	2	FLANGE ADAPTER 300# RF ALLOY 825 B564 UNS N08825 X 12 MM TUBE	0	0	
2660										
2660							***** ORIFICE FLANGES *****			
2670										
2670			ORIFICE FLG	5650830	2	3	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 40S	0	0	
2680	U04		ORIFICE FLG	5623944_C80	4	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2690	U04		ORIFICE FLG	5623947_C80	16	20	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2700	U04		ORIFICE FLG	5623942_C80	24	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 30 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2710	U04		ORIFICE FLG	5623930_C60	4	4	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2720	U04		ORIFICE FLG	5623942_C60	24	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 30 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2730	U04		ORIFICE FLG	5623931_C60	8	8	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 60 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2740	U04		ORIFICE FLG	5623944_C60	10	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2750	U04		ORIFICE FLG	5623947_C60	6	6	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2760	U04		ORIFICE FLG	5623947_C60	16	20	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2770										
2770							***** PIPE NIPPLES *****			
2780										
2780		810		0			NIPPLE PBE CUT FROM PIPE			
2790										
2790		FS1					***** GASKETS *****			
2800										
2800			GASKET	5673657	0.5	24	GASKET 300# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
2810			GASKET	5674034	42	42	GASKET 300# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2820	U04	GA6	GASKET	5673019	2	3	GASKET 300# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.5	0	0	
2830	U04		GASKET	5673700	6	6	GASKET 900# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
2840	U04	FS1	GASKET	5673661	0.5	10	GASKET 600# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
2850										
2850							***** BOLTING *****			
2860										
2860		B1	BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2870										
2870		SB6	BOLTING	5675675	12	42	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2880										
2880							***** BLINDS & SPACERS *****			
2890										
2890			SPEC BLIND	5749683	0.75	3	SPECTACLE BLIND 300# RF ALLOY 825 B424 UNS N08825	0	0	
2900			SPEC BLIND	5749581_C80	4	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2910	U04		SPEC BLIND	5749581_C60	4	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2920										
2920			PADDLE	5750100_C80	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE	0	0	
2930			PADDLE	5750200_C80	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2940			PADDLE	5750028_C80	42	42	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 825 CLAD FACE	0	0	
2950			PADDLE	5750198_C80	42	42	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE	0	0	
2960	U04		PADDLE	5750100_C60	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE	0	0	
2970	U04		PADDLE	5750028_C60	42	42	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 625 CLAD FACE	0	0	
2980	U04		PADDLE	5750200_C60	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
2990	U04		PADDLE	5750198_C60	42	42	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 625 CLAD FACE & BORE	0	0	
3000										

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZE - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 230 °C	12/16/2024	BY OZAC	U04	Page 5 of 5	300C80